

Blood vessel segmentation in retinal images

Abstract

The human eye is one of the most complex organs in the body, and diseases like diabetic retinopathy and glaucoma can silently damage it before any symptoms appear. One of the earliest signs of these conditions shows up in the blood vessels of the retina. Detecting these vessels manually from fundus images is tedious and depends heavily on the expertise of the examiner, which is why an automated approach is needed.

In this project, we built a simple but effective pipeline to automatically segment blood vessels from retinal images using only traditional image processing techniques. We start by isolating the green channel of the image since it naturally highlights vessels the best. From there, we enhance the contrast using CLAHE (Contrast Limited Adaptive Histogram Equalization), extract the vessel regions using a Black-Hat morphological transform, and convert the result into a binary mask using Binary thresholding. A final noise removal step cleans up the output. The outcome is a clear segmentation of the retinal blood vessels, showing that even without deep learning, classical image processing methods can produce meaningful and reliable results.

Objectives

- a) Extract the green channel from the RGB retinal fundus image to maximize vessel-to-background contrast
- b) Enhance local contrast of the green channel using CLAHE to make thin and faint vessels more visible
- c) Apply the Black-Hat morphological transform to isolate dark vessel structures from the bright retinal background
- d) Automatically determine the optimal binarization threshold using Binary thresholding to produce a binary vessel mask
- e) Remove noise from the binary mask using morphological opening to retain only true vessel structures
- f) Display the output of each processing step to visually verify the segmentation pipeline

Assumptions

- a) The input image must be a standard RGB retinal fundus photograph with a circular field of view and a dark surrounding background.
- b) The image must have sufficient brightness and resolution for CLAHE (Contrast Limited Adaptive Histogram Equalization) to effectively enhance vessel contrast.
- c) The Black-Hat kernel size of 15×15 assumes the structuring element is larger than the thickest visible vessel in the image.
- d) Binary thresholding assumes vessel and background pixels are clearly separable, forming a bimodal intensity histogram.
- e) The solution is valid only for healthy or mildly affected retinas severe lesions or haemorrhages may cause false detections.

System Architecture

1) Block diagram

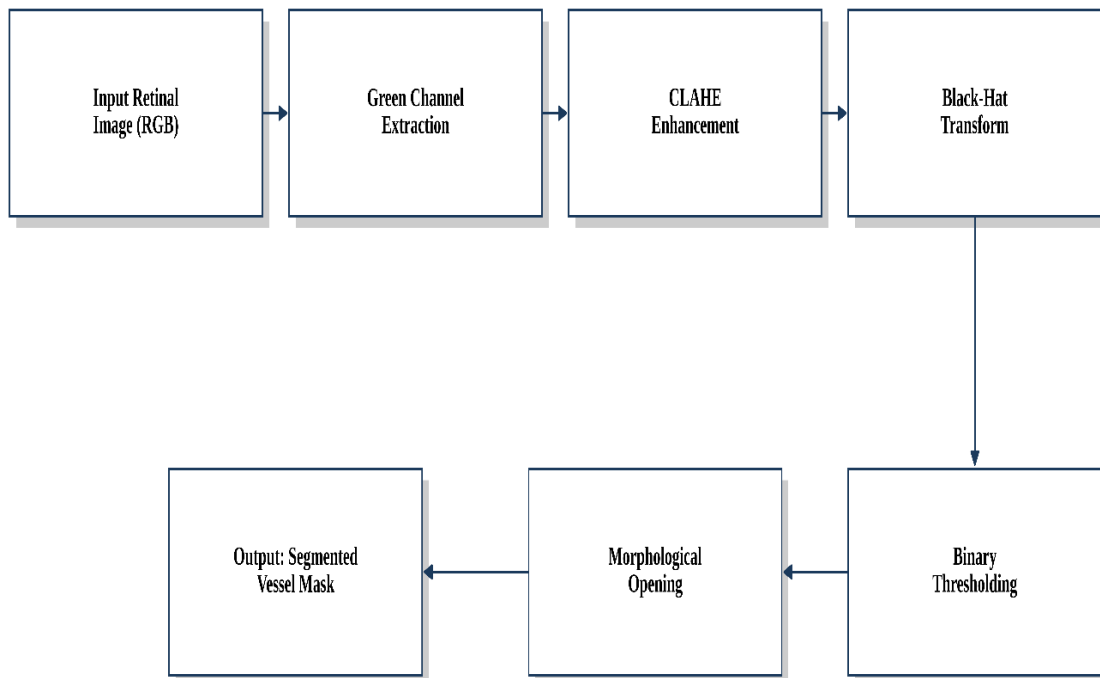


Fig 1: Block Diagram of retinal image

2) Elaborate the Image processing modules

a) **Green Channel Extraction:**

A standard retinal fundus image is captured in RGB format, containing three colour channels Red, Green, and Blue. Among these, the green channel is selected because blood vessels appear darkest against the retinal background in this channel, providing the highest contrast compared to the red and blue channels. The green channel is extracted using channel splitting and used as the grayscale input for all subsequent processing steps.

b) **CLAHE (Contrast Limited Adaptive Histogram Equalization):**

CLAHE is a contrast enhancement technique that operates on small, localized regions of the image called tiles, rather than the entire image at once. Unlike standard histogram equalization which can over-amplify noise, CLAHE limits the contrast amplification using a clip limit parameter, preventing noise from becoming dominant. In this project, CLAHE is applied to the green channel to make faint and thin vessels more visible before feature extraction.

c) **Black-Hat Morphological Transform:**

The Black-Hat transform is a morphological operation that reveals dark structures which are smaller than the structuring element against a bright background. It is computed by subtracting the original image from its morphological closing. An elliptical structuring element of size 15×15 is used because blood vessels are curved and elongated in nature. In this project, the Black-Hat transform effectively highlights the dark vessel regions from the bright retinal background, making them easy to detect in the next step.

d) **Binary Thresholding:**

Binary thresholding converts a grayscale image into a black and white binary image by comparing each pixel value against a threshold. Pixels above the threshold are set to white (255) and pixels below are set to black (0). In this project, the threshold value is automatically determined from the image

histogram so that vessel pixels and background pixels are optimally separated, producing a binary mask where white pixels represent detected vessels.

e) **Morphological Opening (Noise Removal):**

Morphological opening is a two-step operation erosion followed by dilation performed using a small elliptical structuring element of size 3×3 . Erosion removes small white noise spots that are thinner than the structuring element, and the subsequent dilation restores the size of the remaining vessel structures. In this project, this step acts as a post-processing filter to clean up false detections and isolated noise pixels from the binary vessel mask, producing the final clean segmentation output.

Input to the system

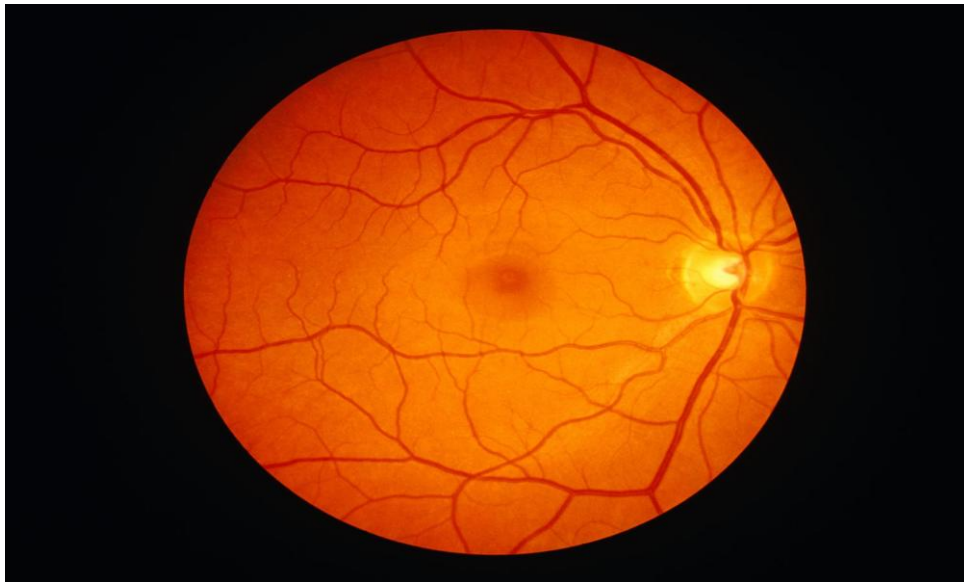


Fig 2: Input of Retinal image

The input to the system is a single retinal fundus photograph sourced directly from Google Images and saved as a JPEG file. The image is a standard colour retinal fundus photograph taken using a fundus camera, showing the interior surface of the eye including the retina, optic disc, and blood vessels. The image is in RGB colour format with a circular retinal field of view surrounded by a black background.

Limitations of Input

- The image must be in RGB colour format grayscale images will not work as green channel extraction requires a 3-channel input
- Heavily blurred or very low-resolution images may reduce segmentation accuracy
- Images with severe retinal diseases such as large haemorrhages or bright lesions may produce false detections

Output

The system processes the retinal fundus image through a 5-step pipeline and displays, the output of each step as a separate window on the screen.

1. **Original** The raw retinal fundus image loaded from the file system. This is the starting point of the pipeline. One should observe a circular retinal disc with visible blood vessels in red/orange tones, a bright optic disc on the right side, and a dark macula at the centre.
2. **Green Channel** The green channel extracted from the RGB image displayed as a grayscale image. One should observe that the blood vessels appear as dark lines against a lighter background this is the highest contrast representation of the vessels among all three colour channels.
3. **Enhanced (CLAHE)** The green channel after applying CLAHE. One should observe that the overall contrast is improved compared to the plain green channel faint and thin vessels that were barely visible before should now appear more clearly defined.
4. **Black-Hat Transform** The result of the Black-Hat morphological operation. One should observe that the image now appears mostly dark with the blood vessel regions appearing as bright white lines the vessels have been effectively separated from the background.
5. **Final Segmented Vessels** The final binary mask after thresholding and noise removal. One should observe a black background with white lines representing the detected blood vessels. The vessel network structure of the retina should be clearly visible with minimal noise or false detections.

Blood Vessel Segmentation in Retinal Images

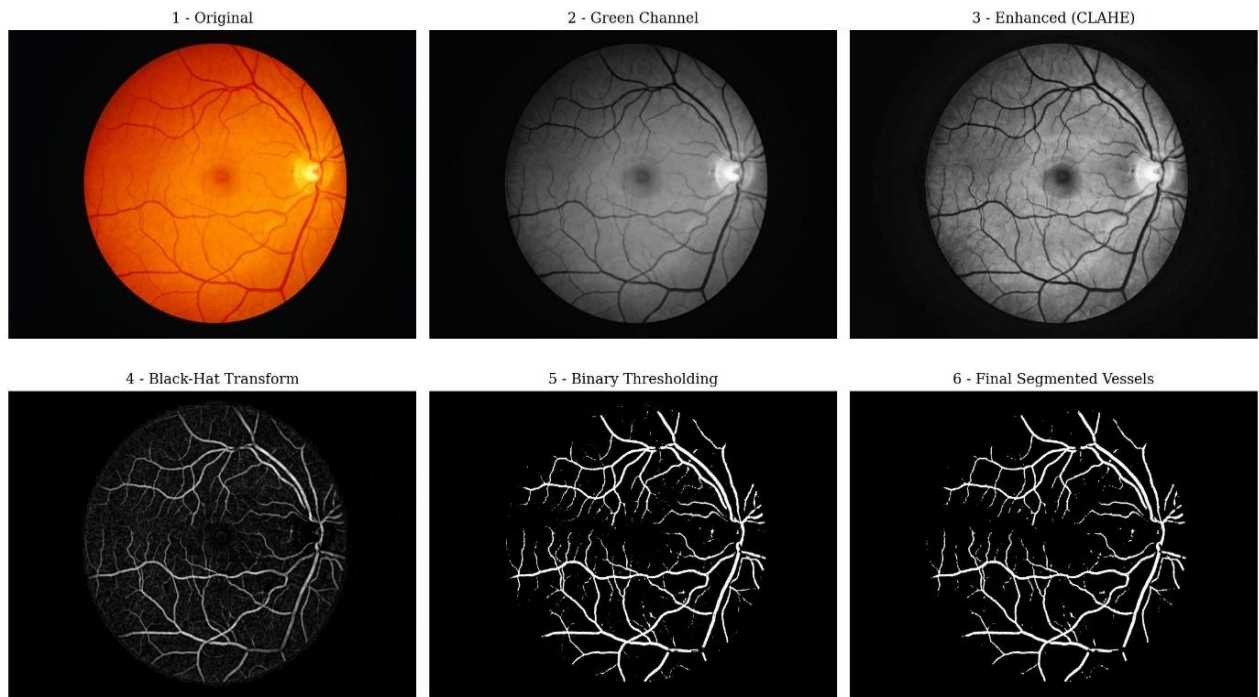


Fig 3: Blood Vessel Segmentation of Retinal image